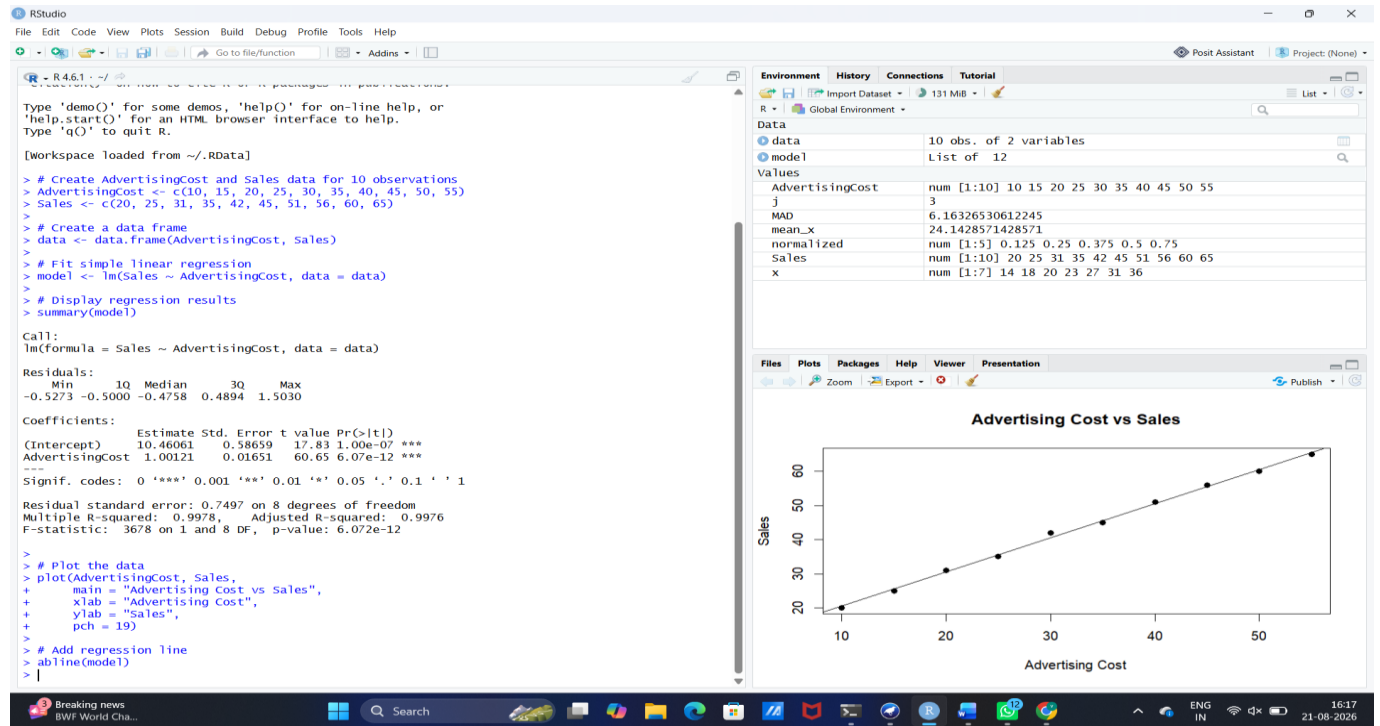
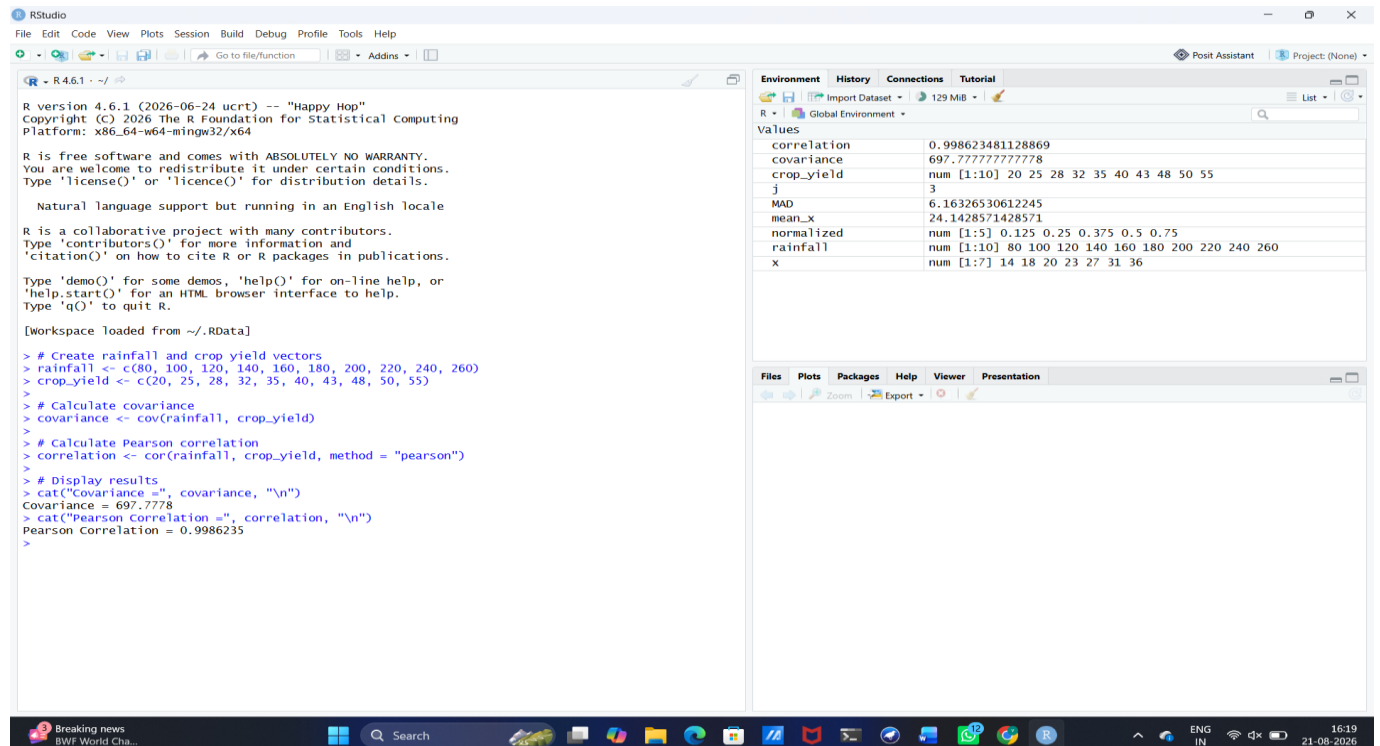


SET 9

1. R: Create AdvertisingCost and Sales for 10 observations; fit simple linear regression and plot the regression line.



2. R: Calculate covariance and Pearson correlation between rainfall and crop yield vectors.



3. WEKA: Create house-price ARFF data (Area, Bedrooms, Bathrooms, Price); apply LinearRegression and report MAE, RMSE and correlation.

Dataset: Missing Values

```
@relation employee_missing
@attribute Age numeric
@attribute Salary numeric
@attribute Department {IT,HR,Sales}
@data
25,35000,IT
30,42000,HR
,48000,IT
35,,Sales
40,60000,HR
28,39000,Sales
,52000,IT
45,70000,HR
```

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The 'Classifier' dropdown is set to 'LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4'. Under 'Test options', 'Cross-validation' is selected with 'Folds' set to 10. The 'Test mode' is '10-fold cross-validation'. The 'Result list' on the left shows '16:27:30 - functions.LinearRegression' selected. The 'Classifier output' pane displays the following information:

```
Scheme: weka.classifiers.functions.LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Relation: house_price
Instances: 10
Attributes: 4
Area
Bedrooms
Bathrooms
Price
Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

Linear Regression Model

Price =

    150.4108 * Area +
    12300.3625 * Bathrooms +
    -24824.1865

Time taken to build model: 0.2 seconds

=== Cross-validation ===
=== Summary ===

Correlation coefficient          0.9962
Mean absolute error             9420.6555
Root mean squared error        11624.3338
Relative absolute error         7.7785 %
Root relative squared error     8.2398 %
Total Number of Instances      10
```

The status bar at the bottom shows 'Status OK' and a 'Log' button.

4. WEKA: Evaluate LinearRegression on the house data using 5-fold and 10-fold CV and compare measures.

Dataset: Attribute Removal

Use the same Employee Missing dataset from Set 10 Q3. Remove Department as the irrelevant attribute after handling missing values.

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseLinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4

Test options

☐ Use training set

☐ Supplied test set

Set...

☒ Cross-validation

Folds5

☐ Percentage split

%66

More options...

(Num) Salary

Start

Stop

Result list (right-click for options)

16:52:39 - functions.LinearRegression

Classifier output

Scheme:weka.classifiers.functions.LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4

Relation:employee_missing-weka.filters.unsupervised.attribute.ReplaceMissingValues-weka.filters.un

Instances:8

Attributes:3

Age

Salary

Department

Test mode:5-fold cross-validation

=== Classifier model (full training set) ===

Linear Regression Model

Salary =

1759.3123 * Age +

-10094.8287

Time taken to build model: 0 seconds

=== Cross-validation ===

=== Summary ===

Correlation coefficient	0.9886
Mean absolute error	1377.9867
Root mean squared error	1612.5059
Relative absolute error	15.121 %
Root relative squared error	14.2816 %
Total Number of Instances	8

StatusOK

Logx 0